

Data Analysis

97 questions · compiled from 2 files

MANIPULATING DATA WITH PANDAS AND NUMPY

1. What does `df.sort_values('score')` do by default?

1. Sorts rows by the score column in ascending order
2. Sorts columns alphabetically
3. Sorts rows in descending order
4. Returns only unique scores

Answer 1 · Sorts rows by the score column in ascending order

MANIPULATING DATA WITH PANDAS AND NUMPY

2. Which argument is used to sort a DataFrame in descending order?

1. `reverse=True`
2. `descending=True`
3. `ascending=False`
4. `order='desc'`

Answer 3 · `ascending=False`

MANIPULATING DATA WITH PANDAS AND NUMPY

3. What is the main purpose of `groupby()` in Pandas?

1. To remove missing values
2. To split data into groups for aggregation or transformation
3. To convert a DataFrame to NumPy
4. To rename columns

Answer 2 · To split data into groups for aggregation or transformation

MANIPULATING DATA WITH PANDAS AND NUMPY

4. What does `df.groupby('team')['points'].mean()` return?

1. The total points for all rows
2. The average points for each team
3. A sorted list of teams
4. The number of teams only

Answer 2 · The average points for each team

MANIPULATING DATA WITH PANDAS AND NUMPY

5. Which function is commonly used after `groupby()` to count rows in each group?

1. `count()`
2. `shape()`
3. `sumrows()`
4. `rows()`

Answer 1 · `count()`

6. What is the difference between count() and size() in grouped data?

1. count() includes missing values, size() excludes them
2. size() includes all rows, count() ignores missing values in each column
3. They are always identical
4. count() works only on Series

Answer 2 · size() includes all rows, count() ignores missing values in each column

7. Which method combines DataFrames vertically by stacking rows?

1. merge()
2. join()
3. concat()
4. append_rows()

Answer 3 · concat()

8. When using pd.concat on rows, what is a common reason to set ignore_index=True?

1. To sort the columns
2. To reset the row index in the combined result
3. To remove duplicates automatically
4. To convert strings to integers

Answer 2 · To reset the row index in the combined result

9. Which operation is most similar to a SQL inner join?

1. pd.merge(left, right, how='inner')
2. pd.concat([left, right])
3. left.append(right)
4. left.sort_values(right)

Answer 1 · pd.merge(left, right, how='inner')

10. What key parameter is typically required when merging two DataFrames on a shared column name?

1. axis
2. on
3. index_only
4. dtype

Answer 2 · on

11. What does axis=0 usually refer to in Pandas operations?

1. Columns
2. Rows
3. Cells
4. Indexes only

Answer 2 · Rows

12. In `df.drop('col', axis=1)`, what does `axis=1` indicate?

1. Drop a row
2. Drop a column
3. Drop missing values
4. Drop duplicates

Answer 2 · Drop a column

13. Which method adds a new column based on existing columns?

1. Using assignment like `df['new'] = ...`
2. `df.addcol()`
3. `df.insertrow()`
4. `df.column_append()`

Answer 1 · Using assignment like `df['new'] = ...`

14. What is the result of `df['a'] + df['b']` when both columns are numeric?

1. String concatenation
2. Element-wise addition
3. A grouped summary
4. A boolean mask

Answer 2 · Element-wise addition

15. Which NumPy feature allows fast arithmetic across entire arrays without explicit Python loops?

1. Recursion
2. Vectorization
3. Serialization
4. Inheritance

Answer 2 · Vectorization

16. What is broadcasting in NumPy?

1. Saving arrays to disk
2. Applying operations to arrays of compatible shapes
3. Sorting arrays by dimension
4. Converting arrays to DataFrames

Answer 2 · Applying operations to arrays of compatible shapes

17. What does `np.arange(0, 5)` produce?

1. `[0,1,2,3,4]`
2. `[1,2,3,4,5]`
3. `[0,5]`
4. Five zeros

Answer 1 · `[0,1,2,3,4]`

18. How does np.arange differ from np.linspace?

1. arange uses a step, linspace uses a specified number of points
2. They are identical
3. linspace only creates integers
4. arange cannot create 1D arrays

Answer 1 · arange uses a step, linspace uses a specified number of points

19. Which NumPy function creates an array filled with ones?

1. np.ones()
2. np.fullones()
3. np.array_one()
4. np.ones_like_list()

Answer 1 · np.ones()

20. What is the shape of np.array([[1,2,3],[4,5,6]])?

1. (3,2)
2. (2,3)
3. (6,)
4. (2,)

Answer 2 · (2,3)

21. Which attribute returns the shape of a NumPy array?

1. ndim
2. shape
3. size
4. dtype

Answer 2 · shape

22. What does arr.ndim return for a 2D array?

1. The number of rows
2. The number of columns
3. The number of dimensions
4. The total number of elements

Answer 3 · The number of dimensions

23. Which expression selects the second row of a 2D NumPy array arr?

1. arr[2]
2. arr[1]
3. arr[:,1]
4. arr[1,1]

Answer 2 · arr[1]

24. What does `arr[:, 1]` select in a 2D NumPy array?

1. All rows from column index 1
2. All columns from row index 1
3. Only the element at row 1 column 1
4. The last column only

Answer 1 · All rows from column index 1

25. Which NumPy function computes the average of array values?

1. `np.sum()`
2. `np.avg()`
3. `np.mean()`
4. `np.middle()`

Answer 3 · `np.mean()`

26. What does `np.sum(arr, axis=0)` do for a 2D array?

1. Sums across columns for each row
2. Sums down rows for each column
3. Returns a single scalar always
4. Removes missing values

Answer 2 · Sums down rows for each column

27. Which Pandas method applies a function along an axis of a DataFrame or Series?

1. `map_axis()`
2. `apply()`
3. `transform_all()`
4. `loop()`

Answer 2 · `apply()`

28. When is vectorized Pandas/NumPy code generally preferred over Python loops?

1. When readability and performance on tabular/numeric data matter
2. Only when data is text
3. Never
4. Only in Jupyter

Answer 1 · When readability and performance on tabular/numeric data matter

29. What is the main goal of communicating results in data analysis?

1. To hide uncertainty
2. To clearly present findings and insights to an audience
3. To avoid visualizations
4. To replace analysis with opinions

Answer 2 · To clearly present findings and insights to an audience

30. Why should visualizations be matched to the question being asked?

1. Because all charts show the same thing
2. Because the wrong chart can obscure or distort the message
3. Because bar charts are always best
4. Because line charts remove outliers

Answer 2 · Because the wrong chart can obscure or distort the message

31. Which chart type is commonly used to show the distribution of a numeric variable?

1. Histogram
2. Pie chart
3. Flowchart
4. Scatter matrix only

Answer 1 · Histogram

32. Which chart is most appropriate for showing the relationship between two numeric variables?

1. Histogram
2. Scatter plot
3. Bar chart
4. Stacked area chart only

Answer 2 · Scatter plot

33. Which chart is often useful for comparing values across categories?

1. Scatter plot
2. Bar chart
3. Heatmap only
4. Line chart

Answer 2 · Bar chart

34. Why can 3D charts be a poor choice in many analytical presentations?

1. They always require NumPy
2. They can make values harder to compare accurately
3. They cannot use colors
4. They only work in notebooks

Answer 2 · They can make values harder to compare accurately

35. What is an important reason to label axes clearly?

1. To increase file size
2. To help the audience interpret the chart correctly
3. To remove the need for titles
4. To guarantee causation

Answer 2 · To help the audience interpret the chart correctly

COMMUNICATING RESULTS

36. Why is it important to mention limitations in your conclusions?

1. To weaken every result
2. To provide honest context about uncertainty and scope
3. To avoid using data
4. To make charts more colorful

Answer 2 · To provide honest context about uncertainty and scope

COMMUNICATING RESULTS

37. What does a misleading y-axis scale potentially do?

1. Improves precision automatically
2. Distorts the perceived magnitude of differences
3. Removes outliers
4. Changes data types

Answer 2 · Distorts the perceived magnitude of differences

COMMUNICATING RESULTS

38. Which practice best improves the clarity of a presentation slide?

1. Adding as much text as possible
2. Using focused visuals and concise takeaways
3. Using multiple unrelated charts on one slide
4. Avoiding titles

Answer 2 · Using focused visuals and concise takeaways

COMMUNICATING RESULTS

39. Why is audience awareness important when presenting analysis?

1. Because technical detail should always be maximized
2. Because different audiences need different levels of detail and framing
3. Because only executives understand charts
4. Because code should replace explanation

Answer 2 · Because different audiences need different levels of detail and framing

COMMUNICATING RESULTS

40. What is the best interpretation of 'actionable insight'?

1. A result that is mathematically complex
2. A finding that can inform a decision or next step
3. Any statistically significant number
4. A chart with many colors

Answer 2 · A finding that can inform a decision or next step

COMMUNICATING RESULTS

41. Which is the best title for a chart in a report?

1. Chart 1
2. Sales
3. Monthly sales increased after the campaign launch
4. Data output

Answer 3 · Monthly sales increased after the campaign launch

42. Why should correlation findings be phrased carefully in presentations?

1. Because correlation proves mechanism
2. Because audiences may incorrectly assume causation
3. Because correlation cannot be visualized
4. Because scatter plots are invalid

Answer 2 · Because audiences may incorrectly assume causation

43. What is a benefit of using annotations in a chart?

1. They can highlight important events or unusual points
2. They automatically clean data
3. They replace legends in all cases
4. They convert charts to tables

Answer 1 · They can highlight important events or unusual points

44. Which approach is best when presenting a complex analysis to non-technical stakeholders?

1. Lead with code and formulas
2. Start with the key takeaway, then support it with simple visuals
3. Avoid conclusions
4. Show every intermediate table first

Answer 2 · Start with the key takeaway, then support it with simple visuals

45. Why might a line chart be preferred for time-series data?

1. It emphasizes changes and trends over ordered time
2. It groups categories by color only
3. It removes missing dates automatically
4. It is always more accurate than tables

Answer 1 · It emphasizes changes and trends over ordered time

46. What is the main risk of overloading a chart with too many categories or colors?

1. The file becomes encrypted
2. The audience may struggle to interpret the message
3. The data becomes incorrect
4. Pandas cannot render it

Answer 2 · The audience may struggle to interpret the message

47. Which of the following best supports reproducible communication of results?

1. Keeping steps undocumented
2. Using notebooks or reports that include code, outputs, and explanations
3. Changing results manually before presenting
4. Deleting intermediate analysis

Answer 2 · Using notebooks or reports that include code, outputs, and explanations

48. What is the strongest conclusion when an analysis shows an association but the study is observational?

1. X definitely causes Y
2. There is evidence of a relationship, but causation is not established
3. The data is useless
4. No interpretation is possible

Answer 2 · There is evidence of a relationship, but causation is not established

49. What is typically the first step in the data analysis process?

1. Building a dashboard
2. Collecting or identifying the data
3. Training a machine learning model
4. Deploying an API

Answer 2 · Collecting or identifying the data

50. Why is asking the right question important in data analysis?

1. It removes the need for cleaning
2. It determines what data and methods are relevant
3. It guarantees causal conclusions
4. It avoids visualization

Answer 2 · It determines what data and methods are relevant

51. Which of the following is the best example of a well-defined analysis question?

1. Tell me everything about sales
2. Why is data useful?
3. How did monthly revenue change after the pricing update in Q2?
4. Can Python solve business problems?

Answer 3 · How did monthly revenue change after the pricing update in Q2?

52. What is the main goal of data wrangling?

1. To create neural networks
2. To clean and transform data into a usable form
3. To replace missing data with random values only
4. To compress files

Answer 2 · To clean and transform data into a usable form

53. Which activity belongs most directly to the exploration phase of analysis?

1. Submitting a project review
2. Summarizing distributions and checking relationships
3. Installing a browser extension
4. Writing only the final conclusion first

Answer 2 · Summarizing distributions and checking relationships

54. What is the purpose of drawing conclusions from data?

1. To make unsupported claims
2. To interpret findings in relation to the original question
3. To avoid using statistics
4. To remove outliers

Answer 2 · To interpret findings in relation to the original question

55. Why should analysts be cautious about inferring causation from correlation?

1. Because correlation always means randomness
2. Because correlated variables may be influenced by confounders
3. Because causation is easier to prove with fewer rows
4. Because correlation only exists in NumPy

Answer 2 · Because correlated variables may be influenced by confounders

56. What is a key advantage of Jupyter Notebooks for data analysis?

1. They only support Java
2. They combine code, output, and narrative text in one place
3. They cannot display charts
4. They prevent package imports

Answer 2 · They combine code, output, and narrative text in one place

57. Which cell type is used to write formatted explanations in a notebook?

1. Code
2. Markdown
3. Raw CSV
4. Terminal

Answer 2 · Markdown

58. What happens when you run notebook cells out of order?

1. The notebook automatically resets variables
2. The kernel state may cause results to depend on previous executions
3. Markdown cells stop rendering
4. Pandas becomes unavailable

Answer 2 · The kernel state may cause results to depend on previous executions

59. What does restarting the notebook kernel do?

1. Deletes the notebook file permanently
2. Clears variables and execution state in memory
3. Exports the notebook to PDF
4. Sorts all cells

Answer 2 · Clears variables and execution state in memory

60. Why is 'Restart & Run All' useful before submission?

1. It checks whether the notebook works from a clean state
2. It converts all code to Markdown
3. It removes visualizations
4. It hides errors

Answer 1 · It checks whether the notebook works from a clean state

61. Which shortcut mode in Jupyter is primarily for editing cell contents?

1. Command mode
2. Edit mode
3. Kernel mode
4. Output mode

Answer 2 · Edit mode

62. If a variable exists in one cell but causes NameError after restart, what is the most likely reason?

1. The variable type changed to string
2. The defining cell was not rerun
3. The notebook lost internet access
4. The Markdown formatting is broken

Answer 2 · The defining cell was not rerun

63. Which Pandas method shows the first few rows of a DataFrame?

1. tail()
2. preview()
3. head()
4. sample_all()

Answer 3 · head()

64. Which attribute gives the dimensions of a DataFrame?

1. dtype
2. shape
3. columns()
4. size()

Answer 2 · shape

65. What does df.info() help you inspect?

1. Only summary statistics
2. Column names, non-null counts, and data types
3. Only missing values in one column
4. Only the first five rows

Answer 2 · Column names, non-null counts, and data types

EXPLORING DATA

66. Which method is most appropriate for summary statistics of numeric columns?

1. describe()
2. info()
3. plot()
4. rename()

Answer 1 · describe()

EXPLORING DATA

67. Why is checking data types early important?

1. Because plotting requires only strings
2. Because incorrect types can break analysis or produce misleading results
3. Because Pandas cannot store mixed types
4. Because NumPy arrays have no types

Answer 2 · Because incorrect types can break analysis or produce misleading results

EXPLORING DATA

68. Which function is commonly used to read CSV data into a DataFrame?

1. pd.open_csv()
2. pd.read_csv()
3. pd.load_table()
4. pd.import_csv()

Answer 2 · pd.read_csv()

EXPLORING DATA

69. What is the main difference between df.shape and df.size?

1. shape returns rows only, size returns columns only
2. shape returns a tuple of dimensions, size returns total number of elements
3. They always return the same value
4. size returns data types

Answer 2 · shape returns a tuple of dimensions, size returns total number of elements

EXPLORING DATA

70. Which method shows the last few rows of a DataFrame?

1. bottom()
2. tail()
3. end()
4. lastrows()

Answer 2 · tail()

EXPLORING DATA

71. What does df.nunique() help identify?

1. The number of missing rows only
2. The number of unique values in each column
3. The number of duplicated DataFrames
4. The number of plots created

Answer 2 · The number of unique values in each column

72. Which method can be used to inspect a random sample of rows?

1. `random_rows()`
2. `choose()`
3. `sample()`
4. `pick()`

Answer 3 · `sample()`

73. Which expression is commonly used to detect missing values in a DataFrame?

1. `df.empty()`
2. `df.isnull()`
3. `df.isnan_only()`
4. `df.missing()`

Answer 2 · `df.isnull()`

74. What does `df.isnull().sum()` usually provide?

1. The sum of all numeric values
2. The count of missing values per column
3. The number of rows after cleaning
4. The average null percentage

Answer 2 · The count of missing values per column

75. When might dropping rows with missing values be risky?

1. When the dataset is too clean
2. When missingness is substantial and may bias results
3. When all columns are integers
4. When using Jupyter

Answer 2 · When missingness is substantial and may bias results

76. Which method removes rows containing missing values by default?

1. `dropna()`
2. `fillna()`
3. `remove_nulls()`
4. `cleanna()`

Answer 1 · `dropna()`

77. What is a common reason to use `fillna()` instead of `dropna()`?

1. To preserve more rows for analysis
2. To convert all columns to strings
3. To sort the DataFrame
4. To remove duplicates

Answer 1 · To preserve more rows for analysis

MISSING DATA

78. Which value is commonly used in Pandas to represent missing numeric data?

1. None only
2. NaN
3. 0 always
4. False

Answer 2 · NaN

MISSING DATA

79. Why can filling missing values with 0 be misleading?

1. Because 0 may represent a real value rather than missingness
2. Because Pandas forbids zeros
3. Because 0 becomes NaN automatically
4. Because it always improves accuracy

Answer 1 · Because 0 may represent a real value rather than missingness

MISSING DATA

80. Which method can replace missing values in a specific column?

1. `df['col'].fillna(value)`
2. `df['col'].dropna(value)`
3. `df['col'].nullreplace(value)`
4. `df.fillcol(value)`

Answer 1 · `df['col'].fillna(value)`

MISSING DATA

81. What should guide your choice of missing-data strategy?

1. The color of the notebook theme
2. The business context and pattern of missingness
3. Always use mean imputation
4. Always drop the entire column

Answer 2 · The business context and pattern of missingness

INDEXING AND SELECTION

82. Which syntax selects a single column from a DataFrame?

1. `df(column)`
2. `df['column']`
3. `df.column()` only
4. `select df column`

Answer 2 · `df['column']`

INDEXING AND SELECTION

83. Which accessor is label-based in Pandas?

1. `iloc`
2. `loc`
3. `iat`
4. `ndim`

Answer 2 · `loc`

84. Which accessor is position-based in Pandas?

1. loc
2. label
3. iloc
4. at

Answer 3 · iloc

85. What does df.loc[3] usually select?

1. The fourth column
2. The row with label 3
3. The third row by position always
4. A random row

Answer 2 · The row with label 3

86. What is the key difference between loc and iloc?

1. loc uses labels, iloc uses integer positions
2. loc is faster and iloc is deprecated
3. iloc works only on Series
4. loc cannot select columns

Answer 1 · loc uses labels, iloc uses integer positions

87. Which expression filters rows where age is greater than 30?

1. df[age > 30]
2. df[df['age'] > 30]
3. df.filter(age > 30)
4. df.loc('age' > 30)

Answer 2 · df[df['age'] > 30]

88. Why are parentheses often important in compound boolean filtering?

1. They improve chart colors
2. They control operator precedence in conditions
3. They convert booleans to strings
4. They avoid importing NumPy

Answer 2 · They control operator precedence in conditions

89. Which operator is used for element-wise AND in Pandas filtering?

1. and
2. &&
3. &
4. AND

Answer 3 · &

90. What is wrong with using `and` instead of `&` for Series comparisons?

1. `and` works only for floats
2. `and` does not perform element-wise comparison on Pandas Series
3. `and` is slower but still correct
4. Nothing is wrong

Answer 2 · `and` does not perform element-wise comparison on Pandas Series

DATA CLEANING

91. Which method removes duplicate rows from a DataFrame?

1. `drop_duplicates()`
2. `unique_rows()`
3. `dedupe()`
4. `remove_repeat()`

Answer 1 · `drop_duplicates()`

DATA CLEANING

92. Why might column names be cleaned or standardized?

1. To make plotting impossible
2. To improve consistency and ease of access
3. To reduce row count
4. To force numeric types

Answer 2 · To improve consistency and ease of access

DATA CLEANING

93. What is a common issue with inconsistent categorical values like 'NY', 'ny', and 'New York'?

1. They are always treated as one category automatically
2. They can split the same concept into multiple categories
3. They crash NumPy arrays
4. They remove missing values

Answer 2 · They can split the same concept into multiple categories

DATA CLEANING

94. Which string method is useful for removing extra spaces from text values?

1. `trimspace()`
2. `strip()`
3. `cut()`
4. `compress()`

Answer 2 · `strip()`

DATA CLEANING

95. Why should you inspect duplicates before dropping them?

1. Because duplicates may represent valid repeated events
2. Because `drop_duplicates` sorts the data
3. Because duplicates cannot be removed from DataFrames
4. Because Jupyter blocks the operation

Answer 1 · Because duplicates may represent valid repeated events

96. Which method is commonly used to rename columns?

1. `rename()`
2. `relabel()`
3. `columns_map()`
4. `setlabels()`

Answer 1 · `rename()`

97. What is the benefit of converting a date column to datetime type?

1. It allows time-based filtering and extraction of date components
2. It removes all missing values
3. It turns dates into categories only
4. It prevents grouping

Answer 1 · It allows time-based filtering and extraction of date components